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Identifying Gut Adaptation and Evolution in Eukaryotes

Background and Question:

The gut microbiome is a complex ecosystem home to thousands of gut microbes, including bacteria, protists, and fungi. Despite being prominent members of the gut microbiome, little work has been done to dissect the evolutionary history and functional role of gut protists compared to their bacterial counterparts. All extant gut eukaryotes descended from free-living relatives that colonized the anaerobic gut ecosystem [1]. These ancestral protists were exposed to the same novel selective pressures, and we hypothesize that, as a result, they underwent similar loss of self-sustaining aerobic metabolic pathways and similar gain of pathways that aid in nutrient acquisition and host interaction driven by convergent evolution, horizontal gene transfer (HGT), or both. We aim to answer the following questions: A) Is there **gain or loss of key metabolic pathways** associated with adaptation to a gut environment in these protists? B) What are the **evolutionary histories of these shared pathways** and gene families across taxonomic groups; are they inherited from a shared ancestor, the product of convergent evolution, or evidence of shared or convergent HGT?

Although the evidence is limited, there are existing observations that support our hypothesis. Convergent loss of aerobic structures is exemplified by the marked reduction or complete loss of mitochondria in gut protists [2]. In the anaerobic environment of the gut, oxidative phosphorylation is no longer necessary, and mitochondria are accordingly diminished or lost. Through systematic metabolic reconstruction, we aim to identify additional pathways shaped by gut adaptation. Convergent HGT has been observed in unpublished lab work and in *Entamoeba*, an Amoebozoan in which over 100 genes were found to have been horizontally acquired from bacteria or archaea [3]. However, no systematic review of HGT has been completed across gut eukaryotes leaving a critical gap in understanding that is addressed in this study.

Many gut protists also harbor gene families present in high copy numbers with unknown functions and limited homology to characterized genes. These genes are poorly annotated, with assignments often restricted to the domain level. Beyond identifying unknown genes and pathways associated with gut adaptation, we are interested in whether any of these lineage-enriched gene families are truly lineage-specific, whether they are associated with gut adaptation, and whether they represent instances of convergent evolution. Known lineage-specific gene families include cadherin-like proteins in Stramenopiles [23] and BSPA-like putative surface proteins in Metamonada and Amoebozoa (Figure 1) [4]. For proteins found broadly across gut protists, such as cysteine proteases/peptidases, we are interested in whether a deeper superclass structure exists. Given their high copy numbers, we predict that these paralogs will be functionally diverse and undergoing rapid turnover. Through clustering and gene tree reconstruction, we aim to characterize gene family acquisition and expansion across gut eukaryotes.

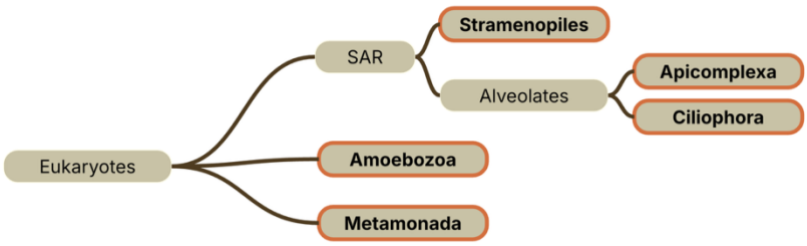


Figure 1: Illustration of protist phylogeny

Methods and Preliminary Results:

Dataset: We used protein sequence data from the following protozoan taxonomy groups: Stramenopiles, Metamonada, Amoebozoa, Ciliophora, Apicomplexa (n=166) (Figure 1). These lineages are comprised of both host-associated and free-living protists where the free-living protists will serve as controls as we identify gut-specific gene families and pathways. Select protein sequence data from Stramenopile species, including *Blastocystis* subtypes, were produced previously in the lab [23]. The remaining sequence data was downloaded from NCBI genome dataset [5]. When selecting from multiple assemblies we picked the reference genome, when possible, followed by the most recently released completed assembly. To support downstream clustering and stratification, we constructed a comprehensive metadata table for all samples using NCBITaxa, incorporating full taxonomic lineages alongside key ecological groupings such as

host-association status (HA vs. FL) and oxygen tolerance (aerobic vs. anaerobic), that are leveraged consistently across all statistical analyses.

Workflow: For efficiency and reproducibility, we created a Snakemake [6] workflow to streamline our analysis. This workflow is currently comprised of four main blocks (Figure 2) which includes 28 rules, 2 checkpoints, and 40 scripts.

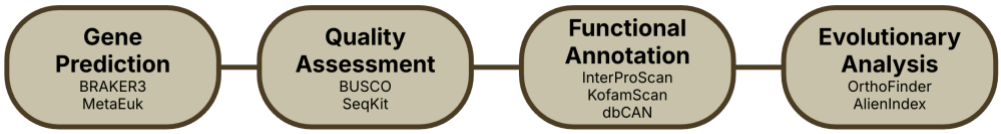


Figure 2: Project Workflow

Gene Prediction and Quality Assessment: While most species had available protein sequence data, several ciliate species lacked protein predictions. To include these species without protein sequence data, we added a gene prediction step to our workflow. Once proteomes were available for all samples, we performed a quality filtering step before proceeding to downstream analyses, using BUSCO [7] to assess relative assembly completeness and SeqKit [8] to filter on minimum sequence length, excluding assemblies that fell below defined thresholds.

Functional Annotation: We then applied three complementary tools for functional gene annotation: KofamScan [9] for KEGG orthology assignment, InterProScan [10] for protein domain and family classification, and dbCAN [11] for carbohydrate-active enzyme (CAZyme) identification. Each tool was run independently on every sample, followed by steps to reduce annotation redundancy and aggregate results into a unified table. We then performed statistical tests, including Log2fc, Mann-Whitney U, and Welch's t-test with Benjamini-Hochberg (FDR) correction to identify genes and corresponding pathways that are enriched and differ significantly across host-association status, aerotolerance, and taxonomic lineage.

Evolutionary Analysis: To investigate the evolutionary origins of identified genes and pathways, we employed OrthoFinder [12] and Alien Index [13]. OrthoFinder was used to infer orthogroups across all samples, enabling us to determine whether shared genes reflect common ancestry or independent acquisition, and to identify lineage-specific gene family expansions. By integrating functional annotation results with orthogroup assignments, we can assess whether genes identified as functionally significant are also linked by shared evolutionary history. AlienIndex, a tool for detecting sequences with anomalous similarity to outgroup taxa, was used to identify putative horizontally transferred genes. We retained regions with high alien index scores that also showed strong hits to other host-associated protists, then merged these regions with InterProScan results to determine whether horizontally acquired genes contribute disproportionately to any functionally significant pathways.

Preliminary Results:

Functional Annotation: KofamScan: We previously identified the steroid biosynthesis pathway as one whose completeness differed significantly by both host-association status and aerotolerance. When we tested module functionality across all KEGG modules in this pathway, we found that free-living and gut-associated anaerobes lacking a functional cholesterol biosynthesis module were missing the same two KEGG orthologs: methylsterol monooxygenase and sterol 14α-demethylase, both of which catalyze oxygen-dependent reactions. The absence of these oxygen-requiring enzymes in nonfunctional species suggests that cholesterol biosynthesis is linked to aerotolerance rather than host-association status. We then built an automated pipeline to examine other primary metabolite pathways with large differences in completion across aerotolerance and host-association groups but found no strong functional patterns among their modules. This was expected, since most primary metabolite pathways are highly conserved. We conclude that loss of cholesterol biosynthesis has been driven by adaptation to anaerobic environment rather than by loss of an easily-scavenged metabolic pathway in host-associated species.

Functional Annotation: InterProScan: Due to small sample sizes for Apicomplexa and Ciliophora, we only had enough statistical power to conduct this analysis on Stramenopiles, Metamonada, and Amoebozoa. We found both domains that were conserved across lineages and domains that underwent independent expansion within each lineage. Each lineage has its own complex cell-surface receptor gene family with functions that are critical for host invasion and immune resistance (Table 1). Domains conserved across lineages include alcohol dehydrogenases, peptidases, glycoside hydrolases, and heat shock proteins. These shared domains all relate to pathogenicity or stress mediation, and because they are found across all lineages, this suggests that they are important for gut adaptation (Table 1).

Evolutionary Analysis: AlienIndex. Using AlienIndex to identify horizontal gene transfer candidates specific to gut adaptation, we again identified regions that mapped to InterPro domains like alcohol dehydrogenases, glycosyl hydrolases, and peptidases. We additionally identified nitroreductases, which also help pathogens withstand harsh conditions such as oxidative stress [14]. Recovering the same domain families through this independent, HGT-based approach corroborates our InterPro enrichment results and strengthens the evidence that horizontal gene transfer, rather than vertical inheritance alone, has shaped gut adaptation across these host-associated lineages.

Evolutionary Analysis: Orthofinder. We used InterProScan results from Blastocystis to build our OrthoFinder pipeline. From the InterProScan results, we identified genes associated with cadherin proteins and filtered OrthoFinder's orthogroups to retain only those that matched these genes. This filtering revealed a large, lineage-specific expansion of orthogroup OG0000108 in Blastocystis. Mapping orthogroups back to InterProScan domains confirmed that OG0000108 corresponds to multiple cadherin family domains we identified in our earlier gene annotation analysis. This orthogroup-level expansion provides independent, gene-family-based support for our earlier finding that each lineage possesses its own expanded complex cell-surface receptor gene family.

Interpro Domain	Lineage	Function
GPCR family 2	Stramenopiles; Blastocystsis (HA)	Surface protein: detect environmental cues, such as nutrient availability or host-derived signaling molecules [15]
Cadherin-like superfamily	Stramenopiles; Blastocystsis (HA)	Surface protein: facilitate host-pathogen attachment and colonization [16]
Giardia variant-specific surface protein	Metamonads; Giardia (HA)	Surface protein: resist degradation by host intestinal proteases [17]
BspA-type LRR region	Metamonads and Amoebozoans	Surface protein: May mediate oxidative stress [18]
Alcohol dehydrogenases	Stramenopiles and Amoebozoans	Provides resistance to oxidative stress in gut bacteria [19]
Peptidases/Proteases	Stramenopiles and Metamonads	Facilitates invasion, nutrient acquisition, and immune evasion [20]
Glycoside hydrolases	Metamonads and Amoebozans	Facilitates invasion, nutrient acquisition, and immune evasion [21]
Heat shock proteins	Amoebozoans and Stramenopiles	Aid in protein folding under stress [22]

Table 3: Key Interpro Domains and function

Proposed Work:

Our preliminary analysis of the Interpro enrichment analysis and AlienIndex analysis has revealed promising targets, but we have not concluded these analyses, and the work remains in progress. To further corroborate our identified InterPro domains, we would like to incorporate results from our dbCAN analysis, which will provide more specific information about enzymatic function. Once all key domains and pathways have been identified, we will conduct phylogenetic-based evolutionary analysis. Our current OrthoFinder results only cover a single gene family and lineage; we would like to expand this into an automated step in our workflow that systematically parses all significant domains identified during gene annotation. For example, even though InterPro and AlienIndex solidified the importance of peptidases/proteases in gut adaptation, we are still interested in their superfamily structure, which will require gene tree reconstruction and phylogenetic analysis. Phylogenetic analyses will provide conclusive answers to which of our candidate horizontal gene transfer events are strongly supported and will resolve between the following possible evolutionary scenarios: whether transfers have occurred from the same bacterial donor separately eukaryotes, separately from different bacterial donors, or once from bacteria to eukaryotes and then transferred between eukaryotes. Phylogenetic analyses will also be used to determine sub-families of shared large gene families, like heat shock proteins and cell surface variant proteins, that are shared between. Lastly, we would like to create a final step in our workflow to aggregate results from all steps as a final output and to begin drafting a manuscript for publication.

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